

NAME:J.PAVANKUMAR

EMAIL:jpk9030@gmail.com

Phno:9030367314

PositionRiskAssistant

1.ARTIFACT

Position Risk Assistant

Problem statement — my take

Open F&O positions can change throughout the day. Prices move, margin requirements can change, and expiry dates get closer. A trader who has several open positions may need to keep checking different numbers to understand whether anything needs attention.

Most trading apps respond to this by showing more information through dashboards, charts, tables, and numbers. While this information is useful, it can still leave the trader with the main question:

What needs my attention right now?

I see this as a prioritization problem rather than a data-display problem. The tool should not simply show the trader their exposure, margin, and expiry information. It should help identify which risk is important enough to look at first and explain the reason in simple language.

The aim is to reduce the time and effort required for a trader to understand the most important risks in their current positions.

Why Nubra should solve it

For an F&O trader, discovering a serious margin issue or an approaching expiry too late can be a frustrating experience. The information may already be available, but the trader still has to identify what is important and decide where to focus.

A tool that highlights the 2–3 most important risks could make this experience simpler. Instead of asking the trader to go through every position and manually compare different numbers, the product can bring the important issues to the front.

I think this can also improve trust in the platform. The product would feel more helpful because it is not only displaying trading information; it is helping the trader understand what deserves attention.

The solution should also avoid making decisions for the trader. It should provide useful context and evidence while leaving the final trading decision with the user.

What is in scope

I am focusing on one main persona:

- A retail F&O trader with around 3–8 open positions who checks their positions during the trading day.

The first version will focus on three risk types:

- **Margin risk** — identify when margin usage is becoming high and may require attention.
- **Expiry risk** — identify positions that are getting close to expiry and may need review.

- **Concentration risk** — identify situations where a large amount of exposure is concentrated in one stock or sector.

The output will be intentionally short. Instead of showing every possible risk, the assistant will provide the **top 2–3 risks**, ranked by priority.

Each risk will contain a simple explanation of what is happening and why it has been highlighted.

What is out of scope

To keep the first version focused and achievable, I will not include:

- Options Greeks or implied volatility analysis.
- Multi-account or multi-broker aggregation.
- Historical backtesting or P&L analysis.
- Live market-price connections.
- Automatic trade execution or position closure.
- Buy/sell recommendations.
- Predictions about future market prices.

The prototype will use static sample data to demonstrate the risk-identification and prioritization logic rather than attempting to build a live trading system.

Solution space

I considered three possible approaches.

1. A full dashboard with all metrics visible

I decided not to use this because it puts the interpretation burden back on the trader. They would still need to search through the information to find the important risk.

2. Show only the single biggest risk

This would make the product very simple, but it could hide another genuine problem. For example, a trader could have both a margin issue and an expiry issue at the same time.

3. Show the top 2–3 risks in priority order

I chose this approach because it provides a balance between simplicity and usefulness. The trader gets a small number of important issues without being overwhelmed by a complete dashboard.

The prototype will therefore take a small set of sample positions, check them for the three selected risk types, rank the important risks, and present them in plain language. The main objective is to help the trader quickly understand **what deserves attention first and why**, without taking the trading decision away from them.

2.ARTIFACT

Prompt 1 — initial build

Build a single-page web app called "Position Risk Assistant" for an options/futures trader.

```
{ "symbol": "NIFTY24500CE", "qty": 50, "entryPrice": 120, "currentPrice": 95, "expiryDate": "2026-09-12", "marginUsed": 45000, "marginAvailable": 48000 },  
  
{ "symbol": "RELIANCE", "qty": 200, "entryPrice": 2850, "currentPrice": 2790, "expiryDate": null, "marginUsed": 0, "marginAvailable": 0 },  
  
{ "symbol": "BANKNIFTY51000PE", "qty": 25, "entryPrice": 210, "currentPrice": 310, "expiryDate": "2026-09-11", "marginUsed": 30000, "marginAvailable": 31000 },  
  
{ "symbol": "TCS", "qty": 150, "entryPrice": 3900, "currentPrice": 3850, "expiryDate": null, "marginUsed": 0, "marginAvailable": 0 },  
  
{ "symbol": "RELIANCE24500CE", "qty": 75, "entryPrice": 40, "currentPrice": 55, "expiryDate": "2026-09-20", "marginUsed": 20000, "marginAvailable": 60000 }  
]
```

Compute three risk types from this data:

1. Margin risk — flag if $\text{marginUsed} / \text{marginAvailable} > 0.9$
2. Expiry risk — flag if expiryDate is within 2 days of today
3. Concentration risk — flag if any single symbol's position value exceeds 40% of total portfolio value

Prompt 2 — refinement

The list currently shows all positions, not just the risky ones. Filter to only positions that trigger one of the three risk rules, then rank by severity (margin risk first, then expiry risk, then concentration risk). Remove any position that doesn't trigger a rule.

Prompt 3 — failed attempt

Make the risk assistant look more professional with a full dashboard showing all metrics for every position.

3.ARTIFACT

Evals — How I decided the output was good enough to ship

I tested the prototype using the sample positions and checked whether it identified the right risks. I mainly looked at three things: whether the risk was detected, whether it was prioritized correctly, and whether the explanation was easy to understand.

Test cases

	What I tested	Expected result	Pass criteria
1	A position where margin used is more than 90% of available margin	Show a margin risk	The position appears in the risk list and the reason clearly explains the margin issue
2	A position expiring within 1–2 days	Show an expiry risk	The position appears and the reason mentions that expiry is close
3	A position making up more than 40% of the portfolio value	Show a concentration risk	The position appears and the reason explains that too much exposure is concentrated there
4	A position with no risk trigger, such as TCS	Do not show it as a risk	The position does not appear in the risk list
5	More than one risk is present	Show the most important risks first	Higher-severity risks are placed higher in the list
6	Fewer than 3 risks are found	Show only the actual risks	The app does not add unnecessary positions just to make three cards

What I considered good enough to ship

For this prototype, I considered the output good enough when:

- Every risk shown could be explained using one of the three rules I defined.
- The app showed a short list instead of turning into a full positions dashboard.
- The reason for each risk was simple enough to understand without looking at the underlying calculations.
- The suggested action gave the trader a clear idea of what they may want to review, without making the trading decision for them.

I did not test live market data, broker integration, or large-scale performance because these are outside the scope of this prototype.

What I would test next

If I had more time, I would test:

- What happens when two risks have the same priority?

- What happens when one position has both margin and expiry risk?
- Whether traders want to tap a risk and see more detail about why it was flagged.
- Whether traders find the suggested actions useful and trustworthy.

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Working App — Position Risk Assistant

LINK: <https://swift-risk-alert.lovable.app>

